

Recall from last class:

- **Theorem 3.3:** If A is an $n \times n$ matrix and E is an elementary matrix, then

$$\det(EA) = \det(E) \det(A),$$

and

$$\det(E) = \begin{cases} 1 & \text{if } E \text{ represents adding a multiple of a row to another row,} \\ -1 & \text{if } E \text{ represents interchanging two rows,} \\ r & \text{if } E \text{ represents scaling a row by a factor of } r. \end{cases}$$

- **Theorem 3.6:** If A and B are n by n matrices, then

$$\det(AB) = \det(A) \det(B).$$

- **Corollary:** A is invertible if and only if $\det(A) \neq 0$.

1. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Let S_1 be the unit square given by $S_1 = \{(x, y) : 0 \leq x, y \leq 1\}$.

(a) Find $T(S_1)$.

(b) How does the area of S_1 related to the area of $T(S_1)$?

(c) Suppose $S_2 = \{(x, y) : 0 \leq x, y \leq 2\}$. How does the area of S_2 related to the area of $T(S_2)$?

2. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Let S be the unit square given by $S = \{(x, y) : 0 \leq x, y \leq 1\}$.

(a) Find $T(S)$.

(b) How does the area of S related to the area of $T(S)$?

Theorem: If A is a 2×2 matrix, then the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, then the volume of the parallelepiped is $|\det A|$.

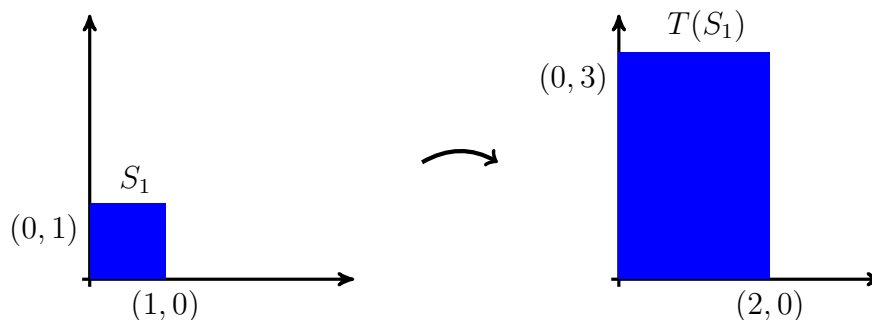
3. Find the area of the parallelogram with vertices $(-1, 0)$, $(0, 5)$, $(1, -4)$, and $(2, 1)$.

Theorem: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear with associated matrix A . If S is a parallelogram in $\mathbb{R}^2$, then $\text{Area}(T(S)) = |\det A| \text{Area}(S)$.

We will generalize this theorem in Math 22b using calculus to more general functions!

Determinant Geometry – Answers / Solutions

1. (a) In the figure below, we see how S_1 transforms under the transformation T .



(b) Notice that $\text{Area}(S_1) = 1$ and $\text{Area}(T(S_1)) = 6$.

(c) Here the area of S_2 is 4, and the area of $T(S_2)$ is 24, so we see that

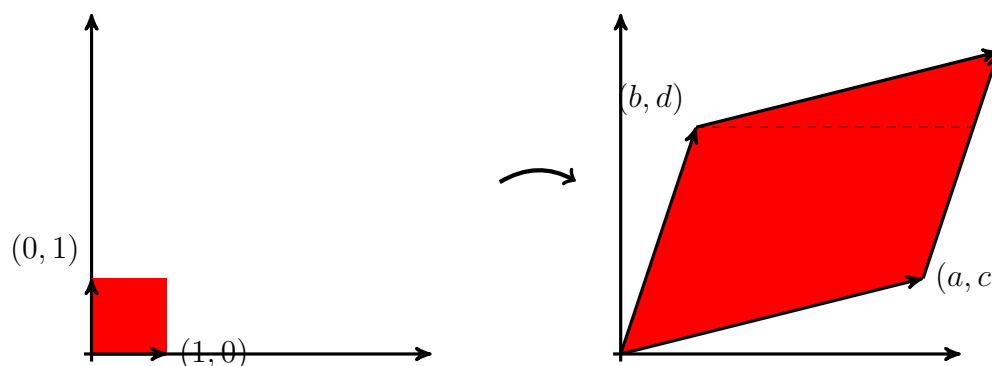
$$\text{Area}(T(S_2)) = 6 \cdot \text{Area}(S_2).$$

In this case, we see that T seems to scale the area of 6, which also happens to be the determinant of the associated matrix. Thus it seems reasonable to guess that

$$\text{Area}(T(S_2)) = |\det A| \cdot \text{Area}(S_2)$$

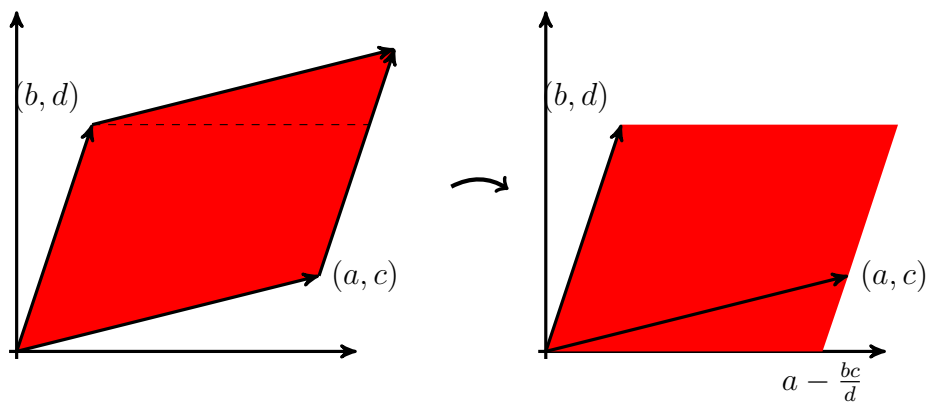
more generally.

2. (a) We show $T(S)$ in the following figure.

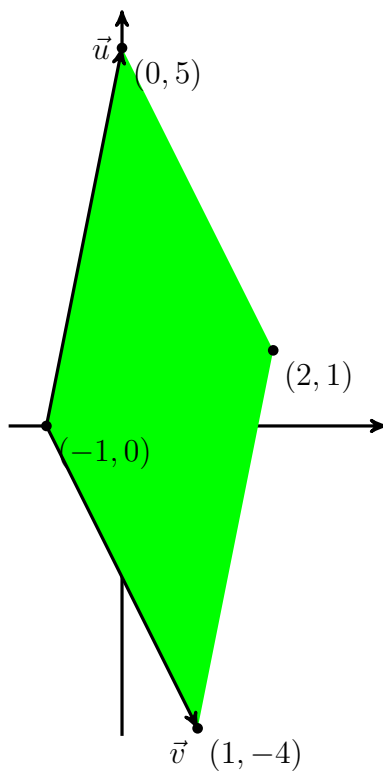


- (b) In the next figure, we show how to rearrange the parallelogram to compute the area. In the parallelogram on the right, we see that the area is given by $d(a - \frac{bc}{d}) = ad - bc = \det A$. Thus we see that

$$\text{Area}(T(S)) = \det A.$$



3. We plot the given points below.



We define $\vec{u}$ and $\vec{v}$ to represent adjacent sides of the parallelogram; namely, let $\vec{u} = \begin{bmatrix} 0 \\ 5 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ -4 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$. Thus the area of parallelogram is given by $\left| \det \begin{bmatrix} 1 & 2 \\ 5 & -4 \end{bmatrix} \right| = |-14| = 14$.